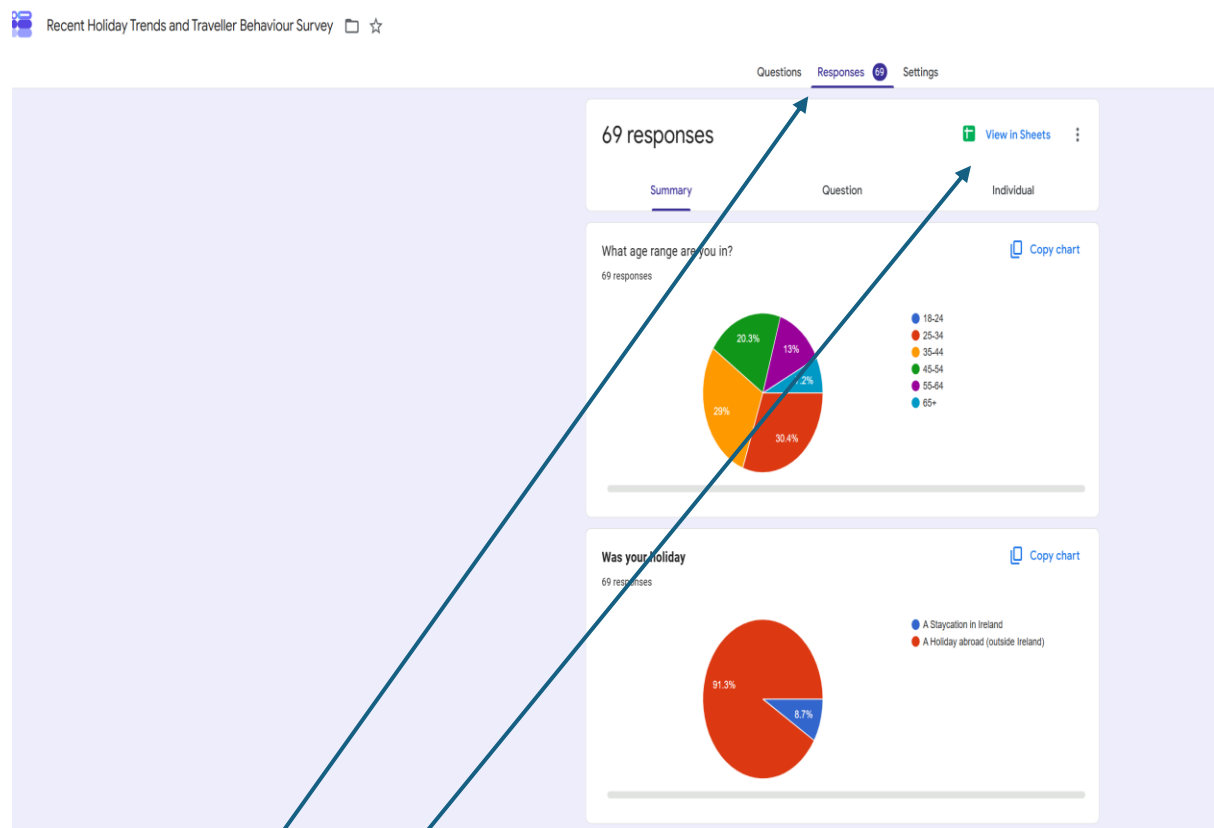


How To Guide for CA1 - Statistical Techniques for DA

Once you have completed your survey on Google Forms. Log in and click on the Responses Tab.

<https://docs.google.com/forms>

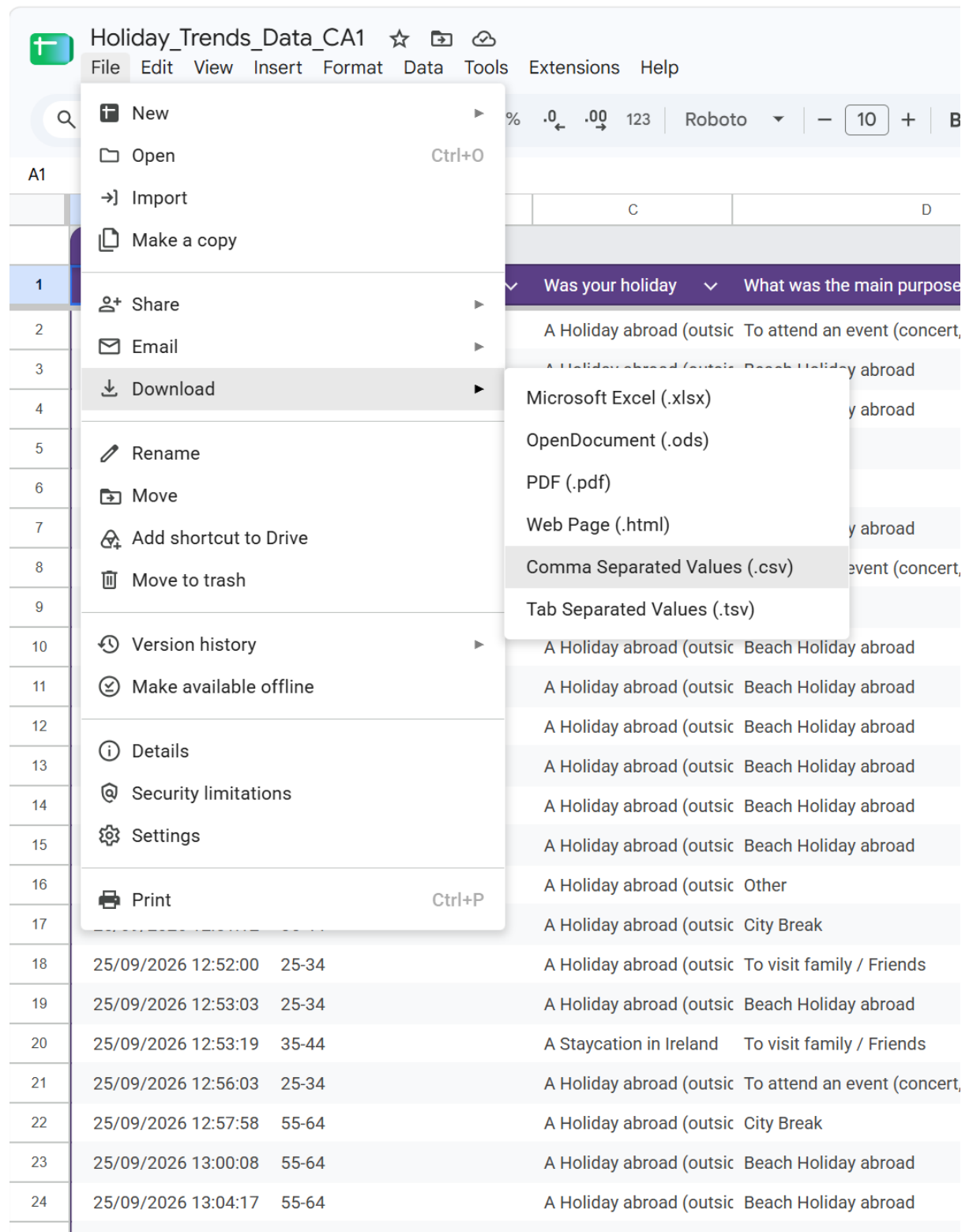


See above

You now want to download your results into a csv format. Click in the link -View in Sheets.

A Google Sheets file will appear. Click on:

File – Download – Comma Separated Values (CSV)

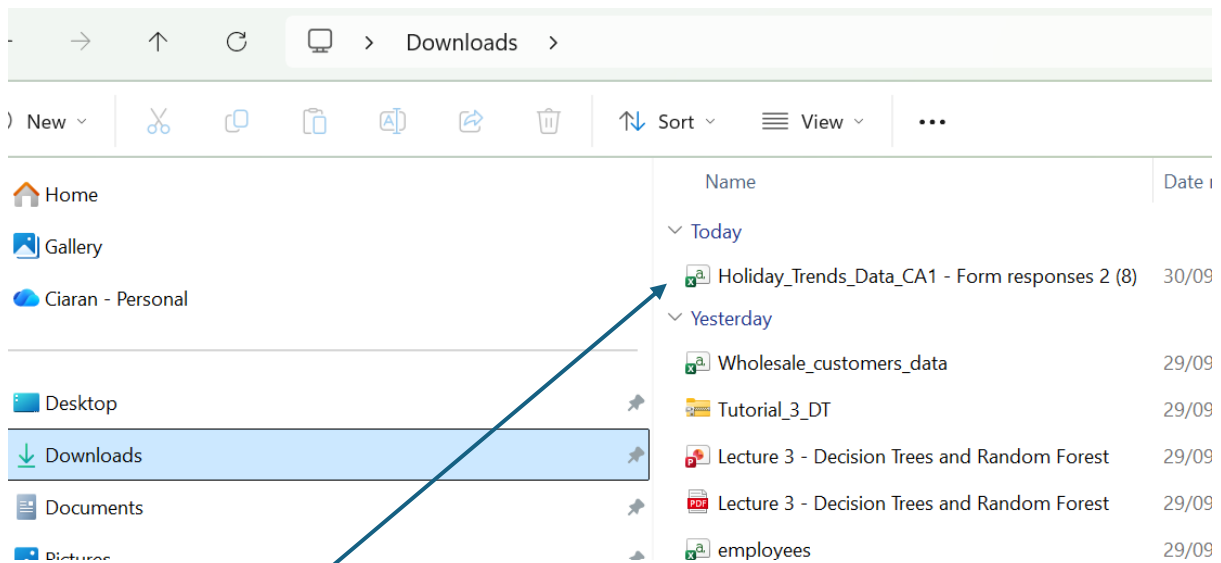


The screenshot shows the Google Sheets interface for a file named "Holiday_Trends_Data_CA1". The "File" menu is open, and the "Download" option is selected. The submenu shows the following options:

- Microsoft Excel (.xlsx)
- OpenDocument (.ods)
- PDF (.pdf)
- Web Page (.html)
- Comma Separated Values (.csv)
- Tab Separated Values (.tsv)

The "Comma Separated Values (.csv)" option is highlighted. The background shows a spreadsheet with columns "Was your holiday" and "What was the main purpose". The data includes various holiday types and purposes, such as "A Holiday abroad (outsic Beach Holiday abroad", "A Holiday abroad (outsic To attend an event (concert", "A Staycation in Ireland To visit family / Friends", and "A Holiday abroad (outsic City Break".

You will see the file in your downloads



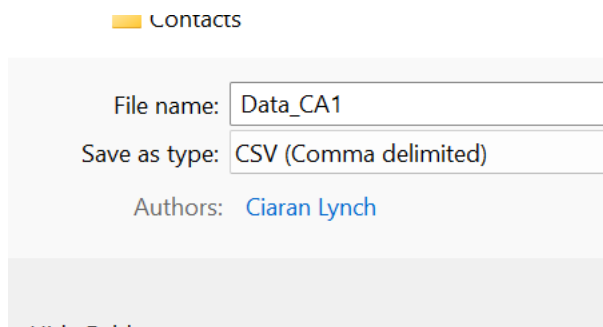
See above

Double click on this file to open then

File - Save As:

Rename to: **Data_CA1** (As per Assignment directions)

Make sure its saving as a CSV



Now make sure that you are putting the file into the folder where **Jupyter Notebook** reads from:

On My pc, Jupyter reads from

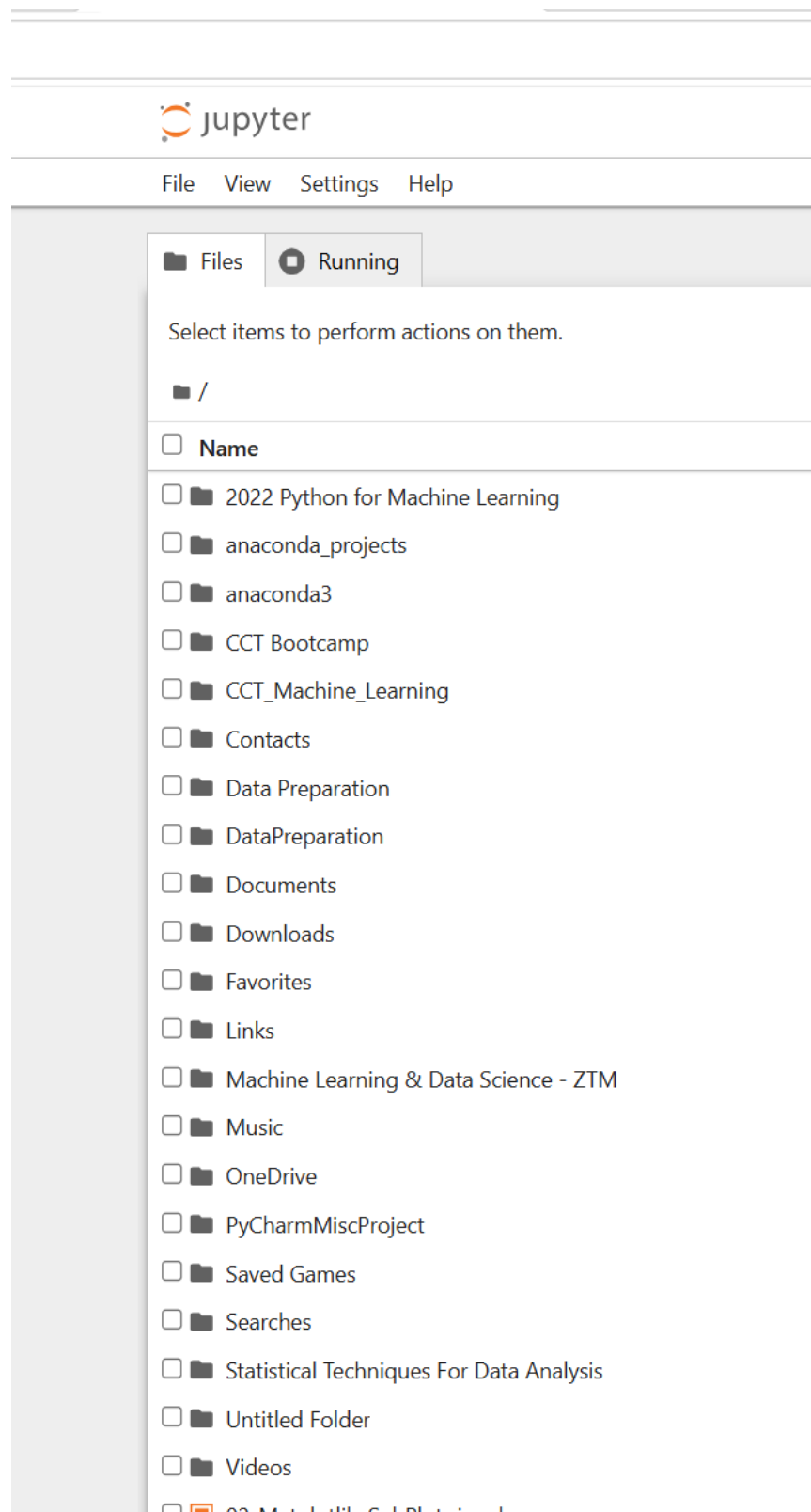
C:\Users\ciara (Thinks I am a woman)

I created a sub Folder called:

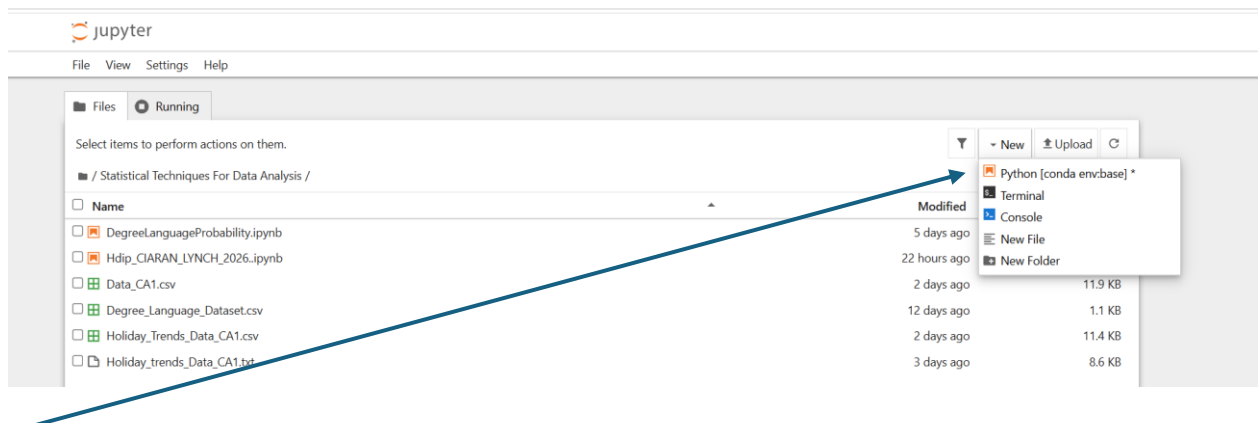
C:\Users\ciara\Statistical Techniques For Data Analysis and saved the file there

Jupyter Notebook

You can now open Jupyter Notebook, via anaconda.



Click on your sub-Folder, if you have one and create a new Python file



Again, as per CA1 directions, I have named the file:

Hdip_CIARAN_LYNCH_2026.ipynb

I gave my Python script a heading.

Type in the first line and use

#Recent Holiday Trends...

Change the file from code to Markdown

Remember

= Heading 1 and ## = Heading 2. ### or ##### tends to be more normal text

Jupyter

Hdip_CIARAN_LYNCH_2026. Last Checkpoint: 2 days ago

File Edit View Run Kernel Settings Help

Python [conda env:base] * Anaconda Toolbox

Recent Holiday Trends and Traveller Behaviour Survey

The purpose of this survey is to explore recent holiday trends and traveller behaviour. This survey forms part of an academic project for the Statistical Techniques for Data Analytics module. [1](#)

Add python libraries to analysis

```
[2]: import pandas as pd
import seaborn as sns
%matplotlib inline
import matplotlib.pyplot as plt
import numpy as np
```

Data Preparation

Load the Dataset

```
[3]: df_Holidays = pd.read_csv("Data_CA1.csv")
```

Data Cleaning

Display first 5 rows, to ensure the data has correctly loaded

```
[4]: df_Holidays.head(5)
```

[4]:

Timestamp	What age range are you in?	Was your holiday	What was the main purpose of your Holiday	How did you book your holiday	How recent was this holiday	How would you describe your holiday type	How many people were in your travel group?	Approximately how much did your accommodation cost per room per night (€)?	How many nights did your holiday last?	Overall Holiday Cost - Approximate total spend for all travellers (€), including cost of holiday + daily spending.	How satisfied were you with your holiday overall?	Would You Take This Type of Holiday Again?	Did You Travel by Air?	How many holidays have you taken in the last 12 months?

Remember:

Step 1 Add the libraries.

Step 2 Load the data

```
df_Holidays = pd.read_csv("Data_CA1.csv")
```

Step 3 Data Preparation

```
df_Holidays.head(5)
```

```
df_Holidays.shape
```

```
df_Holidays.info()
```

```
df_Holidays.isnull().sum()
```

Rename Columns:

```
df_Holidays = df_Holidays.rename(columns={
```

```
"Timestamp": "Timestamp",  
"What age range are you in?": "Age_Range",  
"Was your holiday": "Holiday_Location",  
"What was the main purpose of your Holiday": "Holiday_Purpose",  
"How did you book your holiday": "Booking_Method",  
"How recent was this holiday": "Holiday_Recency",  
"How would you describe your holiday type": "Holiday_Type",  
"How many people were in your travel group?": "Group_Size",  
"Approximately how much did your accommodation cost per room per  
night (€)?": "Accommodation_Cost",  
"How many nights did your holiday last?": "Holiday_Length",  
"Overall Holiday Cost - Approximate total spend for all travellers (€),  
including cost of holiday + daily spending.": "Total_Cost",  
"How satisfied were you with your holiday overall?": "Satisfaction",  
"Would You Take This Type of Holiday Again?": "Holiday_Again",  
"Did You Travel by Air?": "Travel_By_Air",  
"How many holidays have you taken in the last 12 months?":  
"Holidays_Last_12_Months"  
}  
)
```

Once you have finished the Data Preparation, you can move onto:

Descriptive Statistics

Descriptive Statistics

```
[11]: df_Holidays.describe().T
```

	count	mean	std	min	25%	50%	75%	max
Group_Size	66.0	3.333333	1.783112	1.0	2.0	3.0	4.0	10.0
Accommodation_Cost	66.0	206.030303	495.508316	10.0	80.0	117.5	150.0	4000.0
Holiday_Length	66.0	7.257576	5.180765	1.0	4.0	6.0	7.0	28.0
Total_Cost	66.0	2316.666667	1643.947719	200.0	925.0	2000.0	3000.0	6000.0
Satisfaction	66.0	4.454545	0.683471	3.0	4.0	5.0	5.0	5.0
Holidays_Last_12_Months	66.0	3.045455	3.274291	1.0	2.0	2.0	3.0	22.0

Skewness Analysis

```
[12]: df_Holidays.skew(numeric_only=True)
```

```
[12]: Group_Size      1.491254
Accommodation_Cost  7.187998
Holiday_Length     2.518519
Total_Cost         0.724676
Satisfaction       -0.874751
Holidays_Last_12_Months  4.552115
dtype: float64
```

Accommodation Costs shows a positive skewness (6.33), indicating the presence of extreme observations. In Descriptive Statistics, the average accommodation cost is €221.10, the maximum recorded value is €4,000, suggesting a small number of very expensive holidays are influencing the overall distribution

Holiday Length also demonstrates a positive skewness of 2.47. While most holidays show an average of 7.5 days, the maximum holiday length of 28 days was recorded, again this outlier is influencing the overall distribution

- ▼ The Satisfaction skewness of (-0.74) indicates that responses are clustered towards the higher end of the rating scale. This is suggesting that people were generally satisfied with their holiday experience 📊

Median Analysis

```
[13]: df_Holidays.median(numeric_only=True)
```

```
[13]: Group_Size      3.0
Accommodation_Cost 117.5
Holiday_Length     6.0
Total_Cost        2000.0
Satisfaction       5.0
Holidays_Last_12_Months  2.0
dtype: float64
```

See above median values. Accommodation costs has a median of €112.50, however the mean is €221.10. This further demonstrates the impact of extreme values on the distribution

Show counts of dataset for Age Range, Holiday Purpose and Booking Method

```
[14]: df_Holidays['Age_Range'].value_counts()
```

Writing your Report

On the CA1 Document, we are told to create a word report of approx. 1,000 words.

I again have save the word document as:

Hdip_CIARAN_LYNCH_2026.docx

The 3 main areas that we need to discuss are:

Introduction

Analysis & Insights

Conclusion and Outlier Analysis

Create those 3 lines above and make them a Heading1. Then create a table of Contents.

The screenshot shows a word processing software interface. At the top, there is a search bar. Below it, a ribbon menu contains various style options: Normal, No Spacing, Heading (selected), Heading 2, Heading 3, Title, Subtitle, Subtle Empha, and Emp. A blue arrow points from the 'Heading' option in the ribbon to the 'Introduction' heading in the document. The document content includes a table of contents with entries for 'Introduction' (page 0), 'Analysis & Insights' (page 3), and 'Conclusion and Outlier Analysis' (page 4). Below the table of contents, the 'Introduction' section is visible, starting with the text: 'The purpose of this study was to investigate holiday trends and traveller behaviour. As the summer period was just ending, I thought it would be a valid time to complete a survey. With today's cost of living issues, I wanted to gauge if people are still willing to take summer holidays and get an understanding of the overall costs involved.' The document also includes a footer with the text: 'The survey was designed with Google Forms and called: Recent Holiday Trends and Traveller Behaviour Survey'.

Heading	Page
Introduction	0
Analysis & Insights	3
Conclusion and Outlier Analysis	4

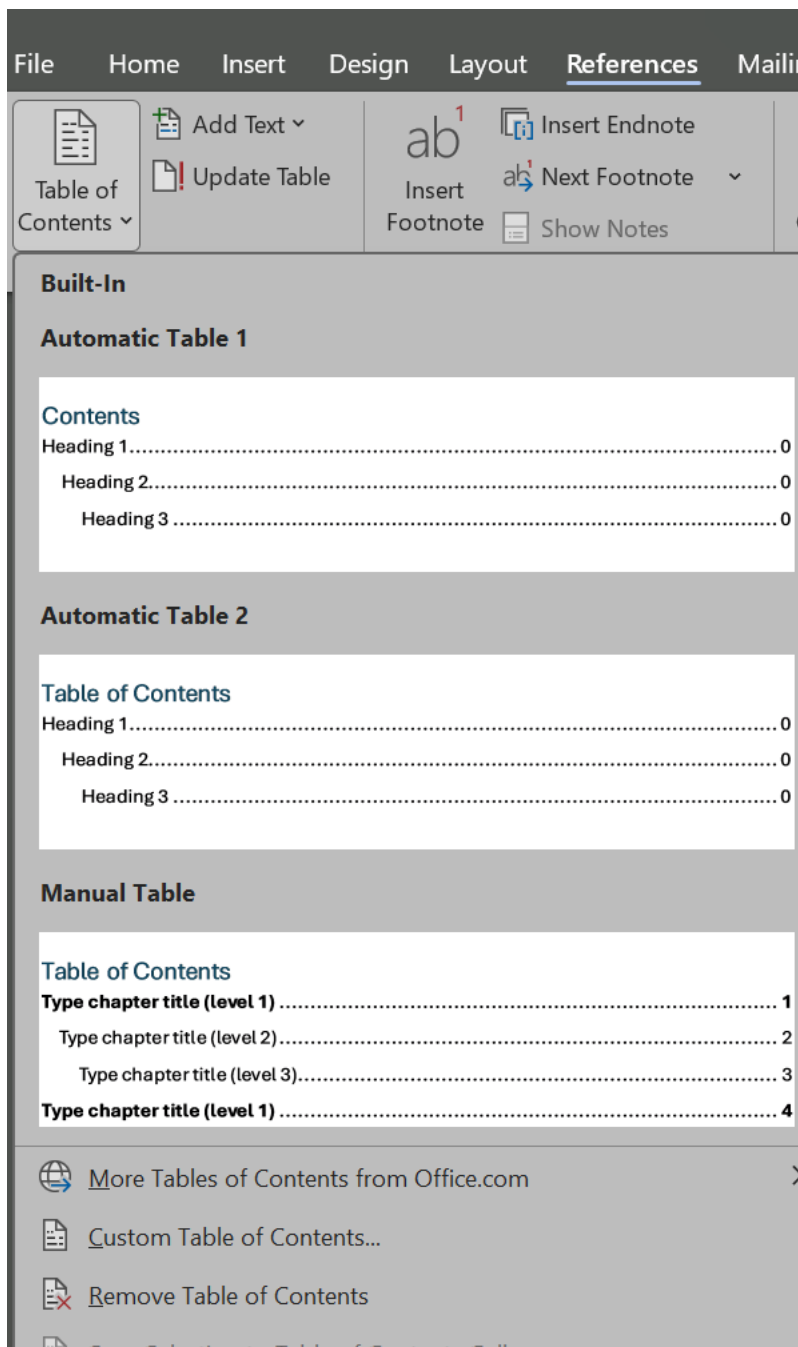
Introduction

The purpose of this study was to investigate holiday trends and traveller behaviour. As the summer period was just ending, I thought it would be a valid time to complete a survey. With today's cost of living issues, I wanted to gauge if people are still willing to take summer holidays and get an understanding of the overall costs involved.

As part of the survey, I wanted to find out if people are still going overseas or preferring to stay in Ireland. Or areas of the survey were to look at the age range of respondents to get an understanding of how they book holidays and how much on average the holiday costs (overall and accommodation costs).

The survey was designed with Google Forms and called:

Recent Holiday Trends and Traveller Behaviour Survey



Downloading your Jupyter Notebook File

To download your Jupyter Notebook, simply select

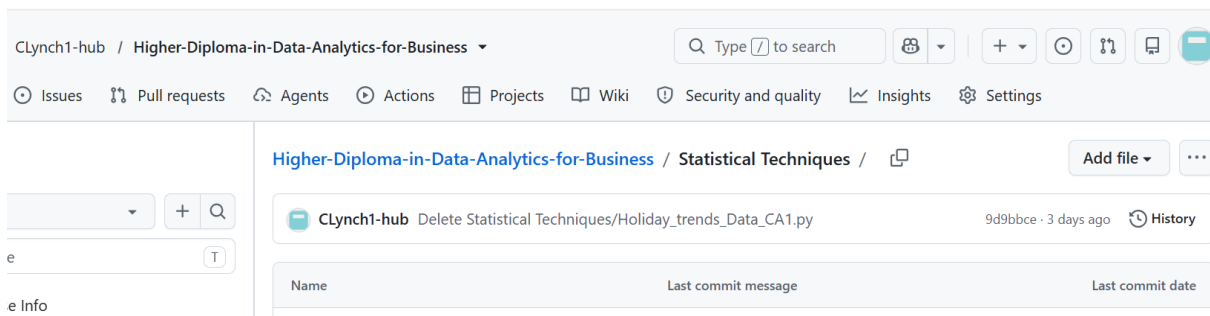
File – Download

Again, this will go to your downloads folder and you can upload it to Moodle.

If you have GitHub set up:

Simply, go to you appropriate Github loation, for this module:

Select Add File – Upload Files



Drag the files from your downloads and the word report from the folder where it is saved and drop them into the add files box.

Once you see the files listed – click Commit Changes

